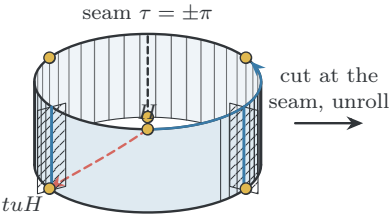
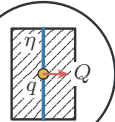
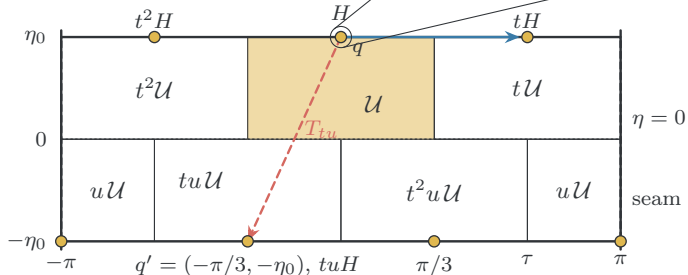


the slice at q :
the cut face of
figure 6, Q across
it



the band of figure 6 with
its six version points, the
two paths, and two normal
slices standing across it



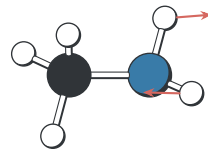
the chart $[-\pi, \pi] \times [-\eta_0, \eta_0]$, seam identified: the six H -cells,
the reference cell $\mathcal{U}$ in gold, the same six points and two
paths; $\eta = 0$ is the planar amino locus; T_{tu} carries q to q'

$$X_0(q)P_{tu} = A_{tu}X_0(q'), \quad A_{tu} = R_z(\pi)$$

$$T_{tu} : (r, \tau, \eta, Q_+, Q_-) \mapsto (rR_z(\pi), \tau - \frac{\pi}{3}, -\eta, Q_+, -Q_-) \quad X = r(X_0(q) + \sum_a Q_a e_a(q))$$

$$M_{tu} = \text{diag}(1, -1) \text{ on the normal directions } (e_+, e_-)$$

$e_{\pm}(q)$ are mass-orthogonal to the family and to the
rotations at q : $\langle e_a, \partial_q X_0 \rangle_{\mathbf{m}} = 0$, $\langle e_a, L_k X_0 \rangle_{\mathbf{m}} = 0$,
 $\langle e_a, e_b \rangle_{\mathbf{m}} = \delta_{ab}$



$e_-(q)$: the amino twist projected onto
the normal space at q ; arrows drawn
 $3 \times$ the displacement $Q_- = 0.42 \text{ \AA}$